

RESEARCH ARTICLE

Generalized Multi-hop Downstream Traffic Pressure for Heterogeneous Perimeter Control

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Abstract

Perimeter control (PC) prevents loss of traffic network capacity due to congestion in urban areas. Homogeneous PC allows all access points to a protected region to have identical permitted inflow. However, homogeneous PC performs poorly when the congestion in the protected region is heterogeneous (e.g., imbalanced demand) since the homogeneous PC does not consider specific traffic conditions around each perimeter intersection. When the protected region has spatially heterogeneous congestion, one needs to modulate the perimeter inflow rate to be higher near low-density regions and vice versa for high-density regions. A naïve approach is to leverage 1-hop traffic pressure to measure traffic conditions around perimeter intersections, but such metrics are too spatially myopic for good PC. To address this issue, we formulate multi-hop downstream pressure grounded in Markov chain theory, which “looks deeper” into the protected region beyond perimeter intersections. In addition, we formulate a two-stage hierarchical control scheme that can leverage this novel multi-hop pressure to redistribute the total permitted inflow provided by a pre-trained deep reinforcement learning homogeneous control policy. Experimental results show that our heterogeneous PC approach leveraging multi-hop pressure significantly outperforms homogeneous PC in scenarios where the origin-destination flows are highly imbalanced with high spatial heterogeneity. Moreover, a sensitivity analysis shows that our approach is robust to turning ratio uncertainty.

KEYWORDS

Traffic pressure, Markov chain, perimeter control

1 | INTRODUCTION

Perimeter control regulates transfer flows among protected regions by limiting the permitted inflow from the feeder links to protected regions, ensuring that each protected region’s vehicle accumulation does not exceed a critical value that would negatively impact traffic productivity in the protected regions. Perimeter control usually relies on Macroscopic Fundamental Diagrams (MFDs), which describe the relationship between traffic productivity and congestion levels and are widely used in numerous studies^{1,2,3}. For MFD-based perimeter control, the design of control parameters relies on a function of the MFD and its critical value of vehicle accumulation^{4,5,6}. However, the accuracy of MFDs is severely affected by the homogeneity of congestion distribution within each region. Lower homogeneity (higher heterogeneity) results in lower reliability of MFDs⁷, and consequently adversely affects the performance of MFD-based perimeter controllers. We clarify at the outset that this paper addresses spatial congestion heterogeneity *within a single protected region*; the complementary problem of partitioning a network into multiple protected regions (reservoirs) and controlling their boundaries is distinct and has been studied elsewhere⁸.

As a concrete example of the limitations of the homogeneity assumption of MFDs, Fig. 1a demonstrates a protected region with internally imbalanced congestion. In this case, choosing a homogeneous inflow around the perimeter may oversaturate the upper subregion while under-utilizing traffic capacity in the lower subregion. This figure is an illustrative snapshot of one heterogeneous-congestion configuration used to motivate the metric; it does not imply that congestion appears instantaneously, and in our experiments the demand and congestion build up and dissipate gradually over time (Section 4.2).

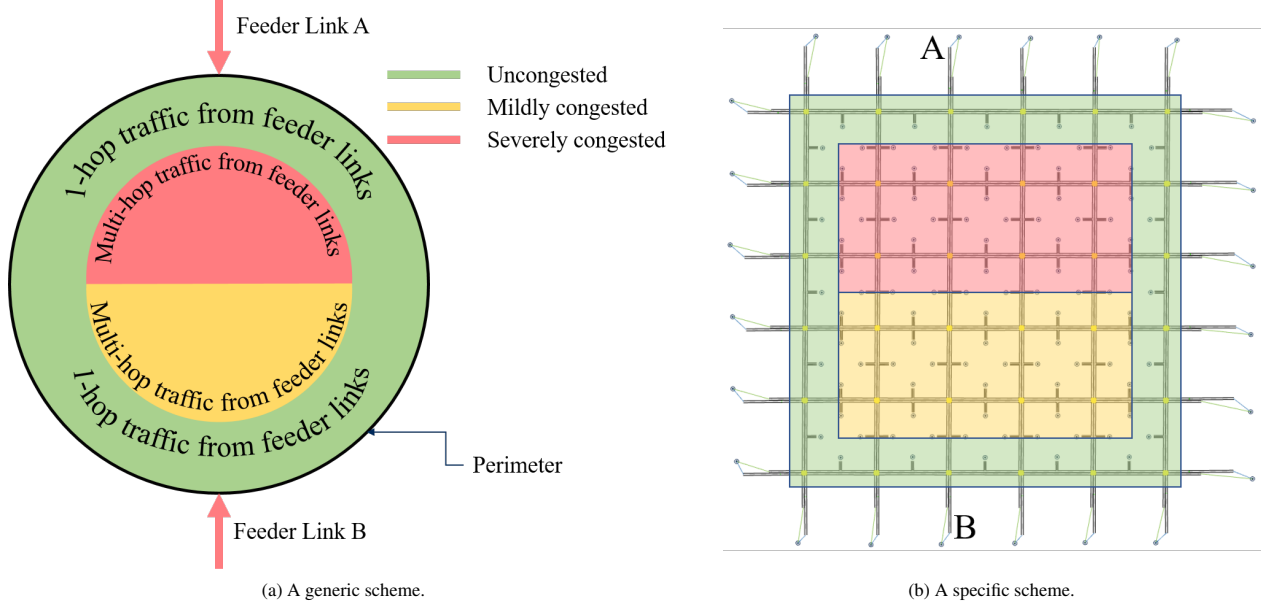


FIGURE 1 Generic and specific schemes showing 1-hop pressure is myopic for perimeter control and the need for multi-hop pressure.

In contrast to MFDs that model aggregated congestion metrics at the region and network level, there is also a class of metrics aiming to measure congestion at the local intersection level. Specifically, the notion of 1-hop traffic pressure quantifies the traffic difference between an upstream link and its immediate downstream links, which is inspired by related work in communication networks^{9,10,11}. Throughout this paper, one *hop* denotes a single transition from a link to a directly connected downstream link, so an h -hop metric reaches links that are h transitions away. Formally, the mathematical definition of 1-hop pressure is in Eq. (1), where $S(l)$ is a traffic statistic (e.g., queue density) for link l , $\mathcal{N}(l, 1)$ is the set of 1-hop downstream links for link l , and T_{lj} is the turning ratio from link l to j :

$$p(l, 1) = S(l) - \sum_{j \in \mathcal{N}(l, 1)} T_{lj} S(j) \quad (1)$$

From the perspective of 1-hop pressure, we can analyze an intersection-level view of congestion in Fig. 1b. The pressure at each perimeter intersection provides a local indication of where to provide perimeter inflow to minimize congestion. In this case, feeder links at the lower perimeter (e.g., feeder link B) have higher pressure than feeder links at the upper subregion (e.g., feeder link A), making it more sensible to permit larger inflows in the lower subregion.

While modulating perimeter inflows according to 1-hop pressure seems to provide a clear solution to the drawbacks of MFD-based homogeneous PC as described above, the existing 1-hop pressure fails to differentiate feeder link A and B, and therefore cannot improve the performance of heterogeneous PC. This creates a need for a metric with *intermediate spatial granularity* that accounts for the gap between the intersection-level metrics (1-hop pressure) and network-level metrics (MFDs) to effectively reflect the nature of heterogeneous PC.

To bridge the gap between the network-level scope of MFDs and the intersection-level scope of 1-hop pressure, we formalize a generalized concept of multi-hop downstream pressure to incorporate a farsighted assessment of protected region congestion. The primary objective is to extend the existing myopic 1-hop traffic pressure metric into a *multi-hop downstream* traffic pressure metric. We emphasize that this metric is a control-oriented spatial measure computed in closed-form, not a model of traffic dynamics: it summarizes downstream congestion into one scalar per feeder link with a customizable spatial horizon, and can be paired modularly with different perimeter-control architectures. Informed by this metric, a heterogeneous perimeter controller can redistribute permitted inflow more effectively based on the specific conditions of each feeder link. Empirically, we demonstrate that a heterogeneous control methodology leveraging our multi-hop generalization of pressure reduces collective travel time compared to the following: homogeneous PC, heterogeneous PC with myopic pressures, and other baseline heterogeneous PC methods under imbalanced protected region congestion.

We summarize the contributions of our work as follows:

1. We provide a mathematical formalization of the multi-hop downstream traffic pressure based on Markov chain theory. This generalization allows us to adjust how far we can reach downstream links, providing a customizable spatial granularity of metrics that bridges the gap between the network-wide scope of MFDs and the single-hop (myopic) scope of the conventional traffic pressure metric;
2. We conduct empirical evaluations with a simple controller leveraging multi-hop pressure for heterogeneous perimeter control under scenarios of various demand heterogeneity;
3. We perform sensitivity analysis that highlights the robustness of our approach against the uncertainties in turning ratio estimations.

2 | LITERATURE REVIEW

2.1 | Perimeter control

Perimeter control manages inter-regional transfer flows by gating the feeder links of protected regions, ensuring that each region's vehicle count does not surpass a critical value that would negatively impact traffic efficiency^{4,5,12,13,14}. Urban traffic networks frequently become oversaturated during rush hour, necessitating perimeter control. The delineation of a congested area is called perimeter identification, which can be executed statically¹⁵ or dynamically^{16,17}. Once the perimeter is identified, perimeter control operates at the network level, and is particularly effective in oversaturated situations where adaptive traffic signal control alone is insufficient¹².

Perimeter control can also be classified as either homogeneous or heterogeneous based on whether the inflow rate is uniformly applied across all access points (homogeneous) or adapted based on local conditions (heterogeneous), both discussed as follows.

2.1.1 | Homogeneous Perimeter Control

Homogeneous PC treats the protected region as a single entity without accounting for local variations around access points. This approach is particularly compatible with macroscopic traffic models (a.k.a. aggregated models or system dynamics models), where individual feeder links are not distinguished. The approaches for homogeneous PC include the following key categories:

- *Feedback Control*: The inflow rate is adjusted to maintain traffic density near a target point, or “set-point” based on the region's MFD. PI controllers are commonly used for this purpose, setting a critical density to optimize traffic flow for the entire region^{4,5,12,18}.
- *Model Predictive Control (MPC)*: MPC techniques¹⁹ have been adapted for homogeneous PC where the control actions are computed by solving the optimization problem over a finite time horizon considering aggregated models, and objectives are related to regional traffic efficiency.
- *Learning-based Control*: Instead of designing controllers based on traffic models, learning-based approaches train a control policy through the interaction with traffic environments. Prior work in²⁰ estimated a pseudo-Jacobian matrix for control from historical data, while²¹ approximated the value function with neural networks for the optimal value function in HJB equations. With deep reinforcement learning (RL), the authors of²² formulated a model-free RL controller for a single protected region PC, then extended it to multiple protected regions²³, but both works are evaluated on aggregated traffic models. Instead, RL was trained on actual traffic networks with microscopic simulation for homogeneous PC in^{24,25}, and²⁴ visualized the contribution of each state variable to the policy.

2.1.2 | Heterogeneous Perimeter Control

Heterogeneous PC allocates different inflows to each access point based on local traffic conditions. Although prior works^{26,27} studied demand heterogeneity with MPC and¹⁷ with a linear quadratic regulator, they only tested on aggregated models instead of actual traffic networks. Therefore, these works do not fall under the category of heterogeneous PC with inflow redistribution around perimeter access points.

The architecture of heterogeneous PC is usually two-stage: the first stage calculates total permitted inflow for the protected region based on homogeneous PC. At the second stage, the total permitted inflow is redistributed based on local traffic conditions.

This approach enables more adaptive control and is particularly beneficial in microscopic simulations that capture detailed network features. The redistribution policies at the second stage can be categorized into the following:

- *Rule-based Redistribution Policies:* These policies distribute total inflow based on simple rules, such as proportional redistribution based on static attributes – saturation flows of feeder links^{4,5,8,28}. This provides a straightforward approach but may not adapt well to dynamic traffic conditions. Other work²⁹ instead integrated the real-time queue length of feeder links with the rule: if low demand or spillback occur, then unused inflows are reallocated to other feeder links. A final approach³⁰ modifies the Q-value of signal phases on perimeter intersections according to the total travel time in protected regions.
- *Optimization-based Redistribution Policies:* Inflow redistribution is framed as an knapsack problem, e.g., one approach⁶ used quadratic programming to redistribute the total permitted inflow from the first-stage PI controller, where the objective is to balance relative queues or delays on feeder links. Another approach³¹ leveraged queues on feeder links to weight the importance of phases for perimeter intersections on the quadratic objective function.
- *Learning-based Redistribution Policies:* Recent studies have applied RL and graph neural networks (GNNs) to heterogeneous control. For example, a notable approach¹³ leverages GNNs to adaptively allocate inflows, learning to prioritize areas with higher demand while imposing penalties for exceeding total inflow limits.

Apart from the two-stage heterogeneous PC above, there are also one-stage heterogeneous PC approaches. Network Max-Pressure (N-MP)¹⁴ combines MaxPressure with heterogeneous PC by considering multi-hop downstream links' queue lengths beyond perimeter intersections. When the average density of the protected region exceeds the critical density, N-MP deprioritizes inflows from those perimeter intersections having higher aggregated congestion level at multi-hop downstream links.

Our work is independently developed and differs from N-MP in several key aspects. Specifically, our approach emphasizes the mathematical formalization of multi-hop downstream pressure using Markov chain theory, which inherently accounts for the decaying influence of higher-hop traffic links. In contrast, N-MP employs an equal weighting scheme that does not incorporate this decaying effect.

2.2 | Macroscopic fundamental diagrams

At the *network level*, Macroscopic Fundamental Diagrams (MFDs) describe the relationship between traffic production and congestion levels for homogeneous urban networks², and have been validated experimentally, e.g., in Yokohama, Japan³². MFDs have been widely applied in studies focused on congestion pricing³³ and traveler information dissemination³⁴. The most notable application of MFDs is in perimeter control, which has been discussed comprehensively in Section 2.1.

Despite their widespread use, MFDs come with inherent limitations. The reliability of MFDs diminishes in regions with high spatial heterogeneity in congestion distribution, which adversely impacts their accuracy^{7,34,35}. Additionally, MFD accuracy can deteriorate if detectors are not ideally placed within the network³⁶. One investigation in urban-highway hybrid networks³⁷ further demonstrated that MFDs can be scattered and heavily influenced by the traffic control plan.

While MFDs are beneficial for high-level strategic traffic management, their broad network-level perspective fails to capture localized traffic conditions around specific areas, such as feeder links. This lack of spatial granularity poses a challenge for applications requiring more detailed, localized control, particularly in addressing the heterogeneous congestion of complex traffic scenarios.

2.3 | Traffic pressure

At the *intersection level*, traffic pressure quantifies a statistical traffic difference (such as vehicle count or density) between upstream and downstream links³⁸, which is inspired by works^{9,10,11} that tackle resource reallocation in wireless communication networks. The application of traffic pressure mainly focuses on the MaxPressure (MP) control policy paradigm, which determines next phase activation without considering phase sequences^{39,40,41,42}, or green time assignment with fixed phase sequences^{31,43,44,45,46} for the next control period. MP methodology has been further incorporated in perimeter control strategies^{31,47}. In a different line of research, traffic pressure has also informed the reward design of RL-based traffic signal control⁴⁸.

There are many variations of definition for 1-hop traffic pressure. In one variant, the link lengths are taken into account⁴³ so that the pressure of a short link with queued vehicles is higher than the pressure of a longer link with the same number of

queued vehicles. In another variant to prioritize some phases, phase weights are proposed^{49,50} in the pressure definition, along with the dynamic estimation of turning ratios. To improve fairness in waiting time, traffic delay has been incorporated into the pressure definition⁴² to permit controllable trade-offs in the relative importance between delay and queues. To avoid the practical challenge of queue measurement, travel times can be used in place of queues⁴⁵ to define the pressure. Similarly, D-MP⁴⁷ used traffic delay to define the pressure, which was empirically shown to be more effective than travel time based MP. To allow large moving platoons to seamlessly travel through an intersection, C-MP⁵¹ proposed to integrate a spatial-mean speed into the pressure definition to prioritize platoon movements. Alternately, OCC-MP⁵² prioritized transit and other high occupancy vehicles in the MP framework to improve passenger-based efficiency. Finally, PQ-MP⁵³ integrated pedestrian queues into the MP framework to address pedestrian traffic.

However, *most existing variations of traffic pressure do not extend beyond immediate downstream links*. While MFDs lack the spatial granularity to capture detailed traffic dynamics, the conventional 1-hop traffic pressure is another extreme that only considers *immediate* downstream links without incorporating further downstream conditions. Although N-MP¹⁴ represents a notable step by integrating multi-hop downstream queue lengths into heterogeneous PC, it applies a uniform weighting scheme that does not account for the decaying influence of distant links.

In contrast, our work formalizes multi-hop downstream pressure through a mathematical framework grounded in Markov chain theory, which inherently incorporates a decaying influence from higher-hop traffic links. This novel metric ensures a more realistic representation of downstream traffic dynamics. Our multi-hop downstream pressure formulation has customizable spatial granularity to address the shortcomings of 1-hop pressure and MFDs. Informed by our multi-hop pressure, we propose a simple yet efficient heterogeneous perimeter controller and test it on scenarios with different levels of heterogeneity. Furthermore, we conduct sensitivity analyses to evaluate the robustness of the proposed controller to turning ratio uncertainty.

We remark that our metric is a strict generalization of the classical pressure-based control: setting $h = 1$ recovers exactly the conventional 1-hop MaxPressure metric (see Eq. (12) with $h = 1$), while $h > 1$ adds an adjustable spatial horizon with turning-probability-weighted decay. We also note a conceptual connection to the traffic-responsive urban control (TUC) strategy⁵⁴: the spatial spread of influence encoded by the powers $\mathbf{P}^h$ is analogous to the off-diagonal structure of the LQR feedback matrix that underlies TUC, and to its “inherent gating” behavior, whereby a network-wide feedback regulator implicitly restrains inflow toward congested subregions⁵⁵. Our metric can be viewed as making this multi-hop, heterogeneity-aware gating effect explicit and directly usable for perimeter inflow redistribution.

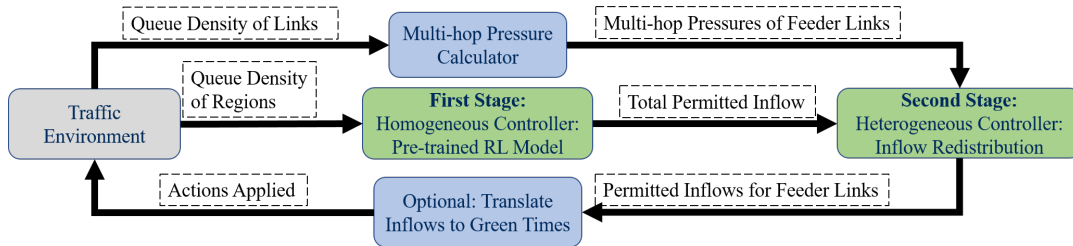


FIGURE 2 The proposed two-stage hierarchical control scheme: the first stage sets the total permitted inflow, which the second stage redistributes across feeder links using multi-hop pressure. Details of each stage and the flow-metering implementation are given in Section 3.2.

3 | METHODOLOGY

We briefly restate the objectives and requirements of our approach. Our aim is to relax the assumptions of traditional MFD-based perimeter control to accommodate our formulation of multihop pressure. At the first stage, we use a model-free deep-RL controller to learn permitted inflow to a PC region independently of any model – MFD or otherwise. At the second stage, we leverage our multihop pressure formulation from each perimeter intersection to heterogeneously modulate the permitted inflow from the first stage on a per-intersection basis. This approach needs only (i) a fixed perimeter with designated feeder links, (ii) observable per-link queue densities, and (iii) turning-ratio estimates, all of which are obtainable from standard traffic sensing. It

does not assume perfect knowledge of network dynamics, further, the sensitivity analysis in Section 6 shows that our approach tolerates substantial turning-ratio error.

An outline of the remainder of this section is as follows: First, we summarize key mathematical notation followed by an overview of our two-stage heterogeneous perimeter control architecture. Then, we present our graph-based representations of traffic networks, highlighting the incorporation of a “supersink” for destination aggregation to achieve the graph’s adjacency matrix to be a Markov chain transition matrix and reduce the size of the graph. Next, we propose the multi-hop pressure metric for evaluating traffic conditions over broader neighborhoods (i.e., multiple links), and we provide mathematical formulations of our multi-hop pressure computation in scalar form for a single link and in vectorized form for all links. Finally, we put all the pieces together in our two-stage architecture: we use multi-hop pressure to inform heterogeneous perimeter inflow rates at the second control stage via a Softmax redistribution of the total permitted inflow from the first control stage.

3.1 | Mathematical notation

Please refer to Table 1 for a summary of the notation that will be used to establish essential concepts for deriving multi-hop pressure and its use in our proposed control architecture. Notation related to experimental setups is introduced in later sections to maintain relevance and coherence for better readability.

TABLE 1 Summary of notation with its description

Notation	Description
a_t^i	Control action for metering device i at timestep t
$Q(l)$	Queue density of link l
$\mathcal{F}$	Set of feeder links
$\mathcal{V}$	Set of all traffic network links (vertices in graph)
$\mathcal{E}$	Set of edges in graph representation
Ω	Supersink
$\mathcal{V}^e$	Extended set of traffic links including supersink
$\mathcal{E}^e$	Extended set of edges
$G = (\mathcal{V}, \mathcal{E})$	Graph representation without supersink
$G^e = (\mathcal{V}^e, \mathcal{E}^e)$	Graph representation with supersink
$\mathcal{N}(l, h)$	Set of h -hop downstream links from l
$p(l, h)$	h -hop pressure for link l
T_{ij}	Turning ratio from link i to j
$\mathbf{T}$	Weighted adjacency matrix of graph G
$\mathbf{P}$	Weighted adjacency matrix of graph G^e , which is also an Markov transition matrix
$\mathbf{Q}$	Concatenation of queue density for links in $\mathcal{V}^e$
$\mathbf{p}(h)$	Concatenation of h -hop pressure

3.2 | Control architecture

The proposed two-stage hierarchical control scheme is shown in Fig. 2. Similar two-stage control schemes are used in heterogeneous PC works mentioned under Section 2.1.

3.2.1 | First stage: computing total permitted inflow leveraging a pre-trained RL control policy

In the first stage, a macroscopic controller computes the total permitted inflow at an aggregated level using macroscopic traffic states (e.g., traffic density of the region) from the traffic environment. This controller is a pre-trained deep reinforcement learning model from our prior research²⁴ that does not rely on MFDs; this model-free RL method exceeds MFD-based baseline methods, making it suitable for our heterogeneous control framework. The visualization of the learned control policy is shown in Fig. 3. In short, higher congestion in the protected region (x-axis) and larger future demand (legend) lead to less permitted inflow (y-axis).

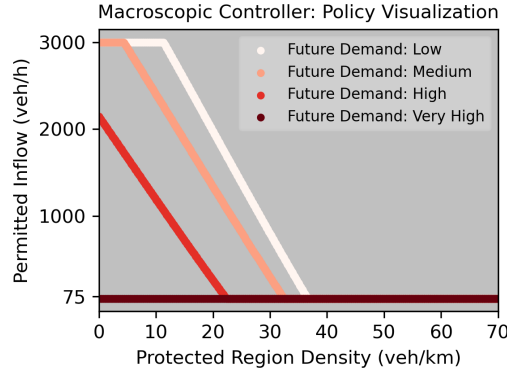


FIGURE 3 The pre-trained control policy for the first stage macroscopic (homogeneous) controller.

3.2.2 | Second stage: redistributing total permitted inflow leveraging multi-hop pressures

In the second stage, the multi-hop pressure calculator computes multi-hop pressures of feeder links around the perimeter using microscopic traffic states (e.g., traffic density of each link). At the end of each control cycle, the total permitted inflow is redistributed by the heterogeneous controller, requiring the multi-hop pressures of feeder links. The controller must satisfy these two requirements:

- *Monotone increasing relationship between pressure and the control action:* a feeder link with a larger pressure should be assigned a larger permitted inflow rate since a larger pressure indicates a potentially higher unused capacity downstream of the feeder link compared to lower pressure feeder links.
- *Total permitted inflow conservation:* The summation of all feeder links' permitted inflows should equal the total permitted inflow provided by the first stage.

Following these requirements, we propose the Softmax controller for its simplicity. We define the *perimeter h-hop pressure vector* $\mathbf{p}_t^{\mathcal{F}}(h) \in \mathbb{R}^{|\mathcal{F}|}$ to be the concatenation of h -hop pressure for each feeder link at timestep t , where all feeder links use the same number of hops (h) in a uniform manner:

$$\mathbf{p}_t^{\mathcal{F}}(h) = \oplus_{f \in \mathcal{F}} p_t(f, h) \quad (2)$$

For feeder link $f \in \mathcal{F}$, its permitted inflow is:

$$a_t^f = A_t \times \text{Softmax}(\mathbf{sp}_t^{\mathcal{F}}(h))_f \quad (3)$$

$$= A_t \times \frac{e^{s \times p_t(f, h)}}{\sum_{f' \in \mathcal{F}} e^{s \times p_t(f', h)}} \quad (4)$$

where A_t is the total permitted inflow at timestep t from the first stage controller, and hyperparameter s (a.k.a. coldness or inverse temperature) is a positive scalar to tune the sensitivity to pressure values. As $s \rightarrow 0$, all permitted inflows are equal and the Softmax controller collapses into homogeneous control. When s is large, a feeder link with higher pressure (over-saturated) is even more prioritized by greater permitted inflow compared to under-saturated feeder links. The choice of s is determined experimentally. By design, the summation of all feeder links' permitted inflows is the total permitted inflow A_t given that the Softmax weights sum to one and hence proportionally redistribute A_t according to the multi-hop pressure.

How inflows are controlled: The inflows are controlled by flow metering instead of signal timings since flow metering provides direct flow control, thus isolating the evaluation of our proposed perimeter control methodology and reducing variability in experimentation. In brief, flow metering restricts the flow of vehicles and releases vehicles with the **precise** amount according to the permitted inflow, while, in contrast, signal timings cannot control the exact number of vehicles to enter. In the traffic simulator AIMSUN⁵⁶, flow metering is implemented with the provided “flow metering” tool.

We stress that this choice does not limit the generality of the method. A permitted inflow a_f (veh/h) can be translated into a green time g_f (s) for the corresponding perimeter phase through $g_f = (a_f/s_f)C$, subject to the usual practical bounds $g_{f,\min} \leq g_f \leq g_{f,\max}$, where s_f is the saturation flow rate of feeder link f and C the cycle length. Thus the redistribution can equivalently be realized through signal green splits on a fixed cycle. We deliberately use flow metering in the controlled experiments so that the reported improvements are attributable to the proposed multi-hop pressure metric itself, rather than to interactions with per-intersection signal-phase design. The qualitative conclusions are expected to persist under signal-timing implementation, however, with additional variability from the conversion of flow constraints to discrete signal timings.

3.3 | Graph representations of the traffic network

We adopt a graph representation to represent the physical traffic network. The graph representation provides a structured way to capture the topological and connectivity characteristics of the traffic network, and therefore we can leverage graph theory for traffic flow analysis. Regarding scalability concerns, since traffic networks have sparse connectivity with each road (link) only directly connecting to neighboring roads (links), the adjacency matrix is also sparse — containing many zeros — which facilitates efficient computation through sparse matrix Python libraries such as SciPy that we use in our implementation. The space complexity of a sparse matrix is $\mathcal{O}(|\mathcal{E}|)$, which scales linearly with the number of edges $|\mathcal{E}|$ in the graph. The time complexity of computing multi-hop pressures depends on the number of hops h and edges $|\mathcal{E}|$: each multiplication with the sparse transition matrix is $\mathcal{O}(|\mathcal{E}|^2)$, and thus computing up to h -hop pressures requires $\mathcal{O}(|\mathcal{E}|^2 \log_2 h)$ time complexity. The $\log_2 h$ factor follows from computing the powers by repeated squaring ($\mathbf{P}^{16} = \mathbf{P}^8 \mathbf{P}^8$, $\mathbf{P}^8 = \mathbf{P}^4 \mathbf{P}^4$, and so on), where the powers $\mathbf{P}^h$ can be pre-computed offline and cached. Empirically, the transition powers remain sparse (non-zeros bounded around 15% of the matrix even at high h). The online per-cycle cost is thus a few sparse matrix-vector products, which is negligible relative to the 96 s control cycle, hence the multi-hop pressure computation can operate comfortably in real-time even for city-scale networks.

We note that the formulation is topology-agnostic: the graph representation and the transition matrix $\mathbf{P}$ are defined purely from link connectivity and turning ratios, with no assumption of a regular grid. It therefore applies without modification to irregular real-world networks; a full field-calibrated validation on such a network is an important direction for future work.

To simplify the graph representation and enable the construction of a Markovian adjacency matrix, we introduce a *supersink* Ω — an abstract node merging all destinations into one — that greatly reduces the number of nodes in graph. The supersink has three key properties:

- *Zero Queue Density*: It has infinite capacity and never rejects vehicles, so $Q(\Omega) = 0$;
- *Absorption*: Any neighbor of the supersink, regardless of hop count, is still the supersink, i.e., $\mathcal{N}(\Omega, h) = \Omega$ for all $h \in \mathbb{N}$;
- *Binary Turning Ratios*: The turning ratio from the supersink to itself or from an exit link (a link solely connected to Ω) to the supersink is 1, and 0 otherwise.

Definition 1 (Graph representation). The graph representation is $G^e = (\mathcal{V}^e, \mathcal{E}^e)$, where:

- The extended link set $\mathcal{V}^e = \mathcal{V} \cup \{\Omega\}$ additionally includes a supersink vertex Ω .
- The extended edge set $\mathcal{E}^e$ additionally includes those edges from exit links to the supersink, and the supersink to itself.
- Edge weight T_{uv} represents the turning ratio from link u to link v . These weights are derived from real empirical data or traffic simulations, encapsulating the probability of traffic flow transitions between links.

To provide a concrete example of this graph representation, a simplistic traffic network with 8 links, shown in Fig. 4a, is mapped onto its graph representation depicted in Fig. 4b. In the graph, each vertex is a traffic link and each edge a connection between links, with edge weights giving the (here fabricated) turning ratios; the supersink Ω and the dashed edges indicate the extension from G to G^e . Note that the toy network in Fig. 4a has no links beyond those illustrated.

3.4 | Vehicle movement as an Absorbing Markov Chain

A vehicle traveling in a traffic network according to certain turning ratios can be formalized as a time-homogeneous absorbing Markov chain by treating a vehicle presenting on a link l as a random variable with probability $Pr(x = l)$. The state space of this Markov chain is a finite set with size $|\mathcal{V}^e|$, which is the total number of traffic links in the entire network. The transition matrix $\mathbf{P}$

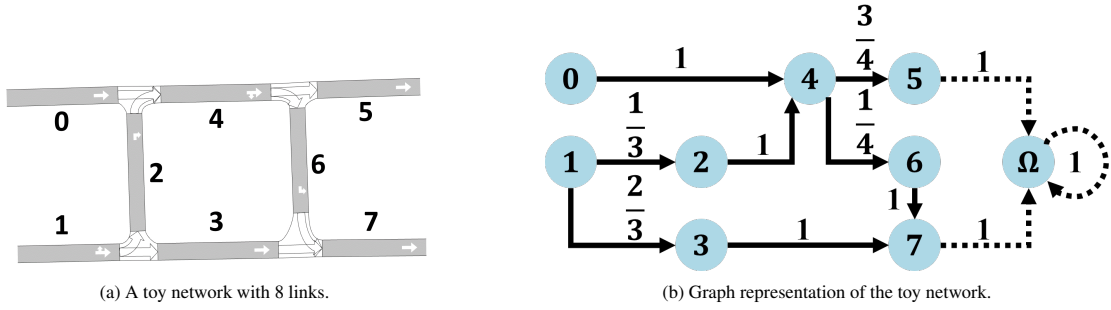


FIGURE 4 Toy network with 8 links (a) and its graph representation (b), used for multi-hop pressure explanation.

is exactly the weighted adjacency matrix of graph $\mathcal{G}^e$, and is defined as follows:

$$\mathbf{P} = \begin{bmatrix} T_{11} & \dots & T_{1|\mathcal{V}|} & T_{1\Omega} \\ \vdots & \ddots & \vdots & \vdots \\ T_{|\mathcal{V}|1} & \dots & T_{|\mathcal{V}||\mathcal{V}|} & T_{|\mathcal{V}|\Omega} \\ 0 & \dots & 0 & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{T} & \mathbf{T}_{*\Omega} \\ \mathbf{0}^\top & 1 \end{bmatrix}, \quad (5)$$

where $\mathbf{T}_{*\Omega}$ is a concatenation of binary turning ratios related to the supersink.

We emphasize that the supersink Ω is an abstract modeling construct in the graph/Markov-chain formulation, not a physical object in the simulator. It aggregates all trip destinations into a single absorbing state so that the weighted adjacency matrix becomes a proper (row-stochastic) absorbing Markov transition matrix, and it also compresses the graph. Operationally, a vehicle “transitioning to Ω ” corresponds to a vehicle reaching its destination and leaving the modeled link set through an exit link; in AIMSUN this is simply trip completion at the destination centroids, and no special simulator object is required.

3.5 | Multi-hop downstream pressure: a customizable metric for long-distance traffic condition

Congestion on a nearby street has a more immediate and significant impact than congestion several blocks away. Therefore, the multi-hop pressure definition needs to capture the diminishing influence of distant congestion while still accounting for its cumulative effect on the current traffic link. We clarify how the influence propagates: it spreads over a *tree of all h -step walks* rooted at the link of interest, not along a single path. Here $\mathcal{N}(l, h)$ denotes the set of links reachable from link l in *exactly* h hops, and each such link contributes with a weight equal to the sum (over all h -step walks reaching it) of the products of the turning ratios along those walks (i.e., the entry of $\mathbf{P}^h$). This is the same accumulated multi-walk influence visualized as the heatmap shading in Fig. 9. To understand the downstream links at different hops, the set of downstream links for link 0 in Fig. 4a at different hops can be written as follows:

$$\mathcal{N}(0, 0) = \{0\} \quad (6)$$

$$\mathcal{N}(0, 1) = \{4\} \quad (7)$$

$$\mathcal{N}(0, 2) = \{5, 6\} \quad (8)$$

$$\mathcal{N}(0, 3) = \{7\} \quad (9)$$

$$\mathcal{N}(0, h) = \{\Omega\}, \quad h \geq 4, h \in \mathbb{N}^+ \quad (10)$$

The scalar version of multi-hop pressure, which computes pressure for a single link, is less compact and therefore included in Appendix A. Unlike the scalar version, the vectorized version can compute pressures for *all* links in the traffic network simultaneously while leveraging sparse matrix operations. Therefore, implementation of the vectorized form accelerates the computation compared to iterating over each link in the network:

$$\mathbf{p}(0) = \mathbf{Q} \quad (11)$$

$$\mathbf{p}(h) = \mathbf{p}(h-1) - \mathbf{P}^h \mathbf{Q}, \quad h \in \mathbb{N}^+ \quad (\text{Recursive form}) \quad (12)$$

$$\mathbf{p}(h) = \mathbf{Q} - \sum_{h'=1}^h \mathbf{P}^{h'} \mathbf{Q}, \quad h \in \mathbb{N}^+ \quad (\text{Unrolled form}) \quad (13)$$

As a base case for validation, we remark that when $h = 1$, Eq. (13) collapses into the vanilla version of traffic pressure:

$$\mathbf{p}(h) = \mathbf{Q} - \mathbf{P}\mathbf{Q} \quad (14)$$

There are two key properties of multi-hop downstream pressure: (1) it is monotone decreasing over the number of hops and (2) it has a bounded range, whose proof can be found in Appendix B.

Traffic-engineering interpretation. In plain terms, the h -hop pressure of a feeder link is its own queue density minus the turning-probability-weighted, distance-decayed sum of the queue densities a vehicle would encounter over the next h links downstream. It therefore measures the *expected downstream spare capacity* that a vehicle admitted at that feeder link will meet: a large positive pressure means the feeder is congested but its downstream is relatively empty (safe to admit more inflow), whereas a large negative pressure means the downstream is congested (admit less). This is a spatial generalization of the store-and-forward back-pressure principle underlying MaxPressure control⁴⁰, now equipped with a tunable look-ahead horizon.

Guarantees and scope. We guarantee key properties required for the use of multi-hop pressure for perimeter control: the pressure is provably monotone in h and bounded to $[-h, 1]$ (Appendix B), and the Softmax redistribution exactly preserves the total permitted inflow supplied by the first stage, so it cannot admit more vehicles than the first-stage macroscopic controller allows. A formal closed-loop stability and convergence analysis of the coupled traffic-controller system would require a tractable dynamical model of the network and closed-form first-stage regulator; since our first stage is a model-free deep-RL policy and our contribution is the metric itself, such an analysis is beyond the present scope. Regarding cyclic routes, because $\mathbf{P}$ is an absorbing Markov transition matrix, the transient block has spectral radius below one, so any closed-walk (looping) contribution is weighted by a product of turning ratios below one and decays geometrically with h , with all probability mass eventually absorbed at Ω . Consequently, cyclic looping cannot artificially inflate the pressure: the metric remains bounded to $[-h, 1]$, and looping contributes only a diminishing correction, which is consistent with the empirical convergence beyond about ten hops.

Interpretations of $\mathbf{P}^h$: It is noteworthy to articulate the term $\mathbf{P}^h$ in Eq. (12). Generally speaking, in the Markov chain, the entry (i, j) in the h -power of transition matrix $\mathbf{P}^h$, denoted as $(\mathbf{P}^h)_{ij}$, represents the probability of starting from vertex i and reaching vertex j after h steps of transitions. In the context of a traffic network where entries represent turning ratios, $(\mathbf{P}^h)_{ij}$ represents the probability of a vehicle moving from link i to link j considering all possible paths of h links:

- *h -hop influence:* The scalar value $(\mathbf{P}^h)_{ij}$ captures the influence that the traffic state of link j has on link i after h transitions, that is, link j is one of the h -hop downstream links from link i , i.e., $j \in \mathcal{N}(i, h)$. It embodies the idea that traffic pressure is not local but can propagate through the network from distant links.
- *Decay of influence over hop:* Since $\mathbf{P}^h$ involves higher powers of $\mathbf{P}$ as h increases, the influence on pressure generally diminishes with hop due to the range of turning ratios being $[0, 1]$. This reflects the realistic attenuation of congestion effects over distance in a traffic network.
- *Pressure contribution:* Multiplying $(\mathbf{P}^h)_{ij}$ by $Q(j)$ incorporates the current queue density of link j into the pressure calculation at link i . If $Q(j)$ is high (indicating congestion), and $(\mathbf{P}^h)_{ij}$ is significant, then link j will contribute substantially to the pressure at link i .

4 | EXPERIMENTAL SETUP

The proposed perimeter control scheme is evaluated using the traffic simulator AIMSUN⁵⁶, which is widely accepted by research institutions, government transportation agencies, and consultancies. Detailed description of the experimental setup in traffic network architecture and the traffic demand are provided in this section.

4.1 | Traffic network architecture

To simulate a monocentric city during peak hours, we created a grid network with 36 signalized intersections that use a fixed control plan, where the signal timings and movements in each phase are provided in Fig. 5 along with link channelization. Similar grid networks are used in^{13,14}. In addition, there are 4 interphases with 4 seconds duration each. Hence, the cycle length is $10+30+30+10+4 \times 4 = 96$ seconds. To be consistent, the perimeter control action update cycle is 96 seconds as well. There are 24 feeder links around the perimeter indexed by integers. Each traffic link is 85 meters long (each block is $2 \times 85 = 170$ meters long) with two lanes per link and the link channelization is identical for all links, except those single-lane links (mimicking parking lot ramps) in the protected region connected to origins or destinations. Inflow metering is used for feeder links as it provides precise control over inflow without disrupting the internal signal plan. As shown in Fig. 5, the protected region is bounded by a red rectangle (the perimeter) and is split into two equal-sized upper and lower subregions, used to generate imbalanced demand.

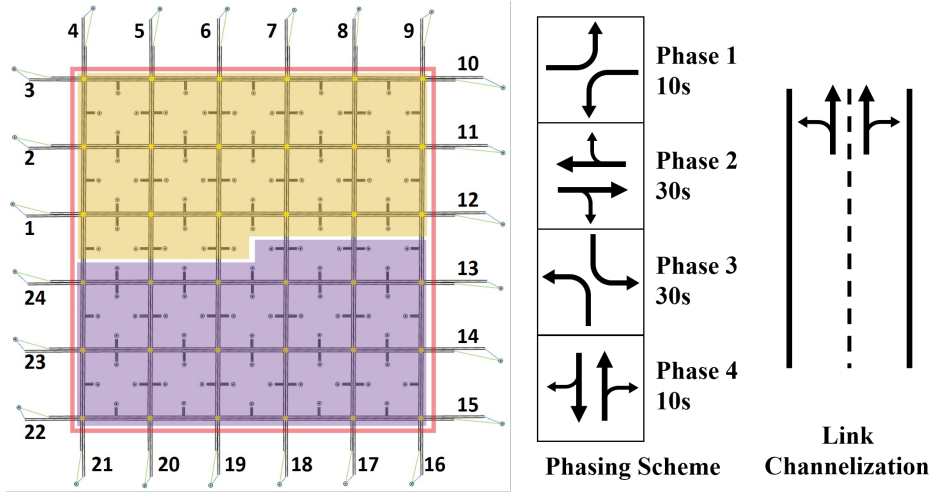


FIGURE 5 Tested grid traffic network; the red rectangle marks the perimeter, and the yellow and purple areas are the upper and lower subregions.

4.2 | Heterogeneous demand setup

We address a key question in this research: does the relative performance improvement of our two-stage multi-hop pressure perimeter control increase as the demand heterogeneity increases? To answer this research question (indexed as RQ3 in later sections), we design scenarios with different levels of demand heterogeneity. Our experiment examines two distinct approaches to modulating demand heterogeneity:

1. *Asynchrony* τ (temporal heterogeneity): the time difference between upper and lower subregions' peak demands. As asynchrony increases, we expect a rise in congestion heterogeneity. In reality, τ can be estimated from either traffic sensor data or simulation.
2. *Imbalance* $\alpha_{22}^{\text{upper}}$ (spatial volume heterogeneity): the difference of demand volume magnitude generated in the upper and lower subregions. As the imbalance increases, we expect a rise in congestion heterogeneity.

To formalize the demand setup, we define mathematical notations in Table 2. The mathematical formulation of demand is as follows:

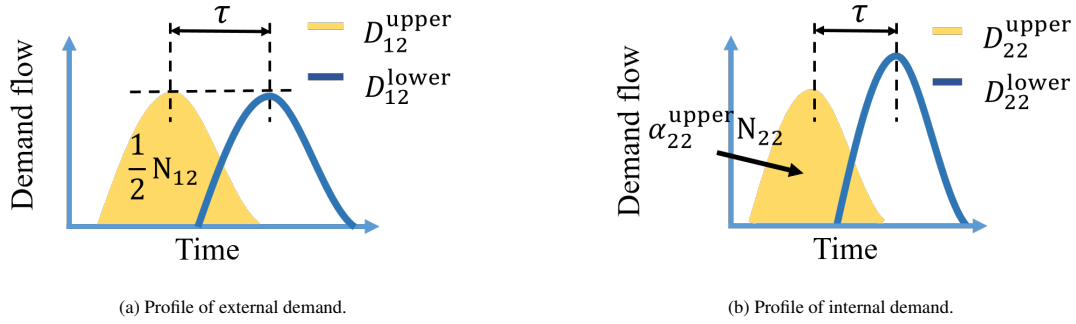


FIGURE 6 Internal and external demand profiles.

TABLE 2 Summary of notations for traffic demand

Notation	Description
T_S	Total traffic simulation duration
N_{12}	Total number of external trips
N_{22}	Total number of internal trips
$D_{22}^{\text{upper}}(t)$	Upper subregion internal demand at time t
$D_{22}^{\text{lower}}(t)$	Lower subregion internal demand at time t
$\alpha_{22}^{\text{upper}}$	Upper subregion internal demand percentage
$\alpha_{22}^{\text{lower}}$	Lower subregion internal demand percentage
$D_{12}^{\text{upper}}(t)$	Demand from upper feeder links to upper subregion at time t
$D_{12}^{\text{lower}}(t)$	Demand from lower feeder links to lower subregion at time t
τ	Asynchrony parameter (time shift)

$$\int_0^{T_S} (D_{12}^{\text{lower}}(t) + D_{12}^{\text{upper}}(t)) dt = N_{12} \quad (15)$$

$$\int_0^{T_S} (D_{22}^{\text{lower}}(t) + D_{22}^{\text{upper}}(t)) dt = N_{22} \quad (16)$$

$$D_{12}^{\text{lower}}(t + \tau) = D_{12}^{\text{upper}}(t), \quad \forall t \quad (17)$$

$$\frac{D_{22}^{\text{lower}}(t + \tau)}{\alpha_{22}^{\text{lower}}} = \frac{D_{22}^{\text{upper}}(t)}{\alpha_{22}^{\text{upper}}}, \quad \forall t \quad (18)$$

$$\alpha_{22}^{\text{upper}} + \alpha_{22}^{\text{lower}} = 1 \quad (19)$$

$$0 < \alpha_{22}^{\text{upper}}, \alpha_{22}^{\text{lower}} < 1 \quad (20)$$

Eq. (15) and Eq. (16) indicate the total number of trips for external and internal demands. Eq. (17) describes the asynchrony between demands from upper and lower feeder links, as illustrated in Fig. 6a. Eq. (18) describes the asynchrony and imbalance of internal demand, as illustrated in Fig. 6b. The asynchrony is regulated by the parameter τ , where a greater τ indicates later internal and external demands at the lower part of the traffic network. For imbalance, we adjust the parameter $\alpha_{22}^{\text{upper}}$ to vary the percentage of internal demand assigned to the upper subregion. For example, $\alpha_{22}^{\text{upper}} = 50\%$ means equal volume of upper and lower subregion demands. As $|\alpha_{22}^{\text{upper}} - 50\%|$ increases, we expect a rise in congestion heterogeneity. Eq. (19) and Eq. (20) ensure the range for parameters related to imbalance. For simplification, there are no trips from the protected region to feeder links, or trips from feeder links to feeder links.

4.2.1 | Tested demands

The tested demands are designed to address the following research questions (RQs) regarding the performance of our proposed two-stage multi-hop pressure perimeter control methodology:

- RQ1: How does the pressure sensitivity s impact the performance of the heterogeneous controller?
- RQ2: What is the optimal range of hops for the best performance of the heterogeneous controller?
- RQ3: Does the performance improvement increase as the demand heterogeneity rises for heterogeneous control?

For RQ1 and RQ2, the demand is set with $\tau = \frac{3}{4}$ hour and $\alpha_{22}^{\text{upper}} = 50\%$. Hence the source of demand heterogeneity comes from asynchrony, not from imbalance. The sensitivity parameter s is tested with values ranging from 2^0 to 2^7 , while the number of hops is varied from 0 to 22 in increments of 2 (each block in the traffic network requires 2 links to cross). Table 3 shows the enumeration of parameter s and h . This results in $8 \times 12 = 96$ combinations of s and hop. The two hyperparameters s and h are therefore selected by grid search over this enumeration, choosing the combination that minimizes the TTS averaged over 10 random seeds; this is precisely the study reported for RQ1 and RQ2.

For RQ3, the optimal controller determined by the best combination of s and hops h based on minimal total time spent (TTS) is tested across various values of τ and $\alpha_{22}^{\text{upper}}$ that determines the demand heterogeneity. The τ values are 0, $\frac{1}{4}$, $\frac{2}{4}$, $\frac{3}{4}$, 1, and the $\alpha_{22}^{\text{upper}}$ values range from 50% to 80% in 5% increments, enumerated in Table 3. This results in $5 \times 7 = 35$ combinations of τ and $\alpha_{22}^{\text{upper}}$.

Both internal and external demands are divided into nine 15-minute intervals, and the numbers of trips for these nine intervals are determined by a vector of nine weights: $[r^0, r^1, r^2, r^3, r^4, r^3, r^2, r^1, r^0]$, where r (fixed to 2) serves as a hyperparameter to regulate the peakedness of the demand profile. This rising-then-falling profile means the demand, and therefore the congestion, builds up progressively to a peak and then dissipates over the simulation horizon rather than appearing instantaneously; the controller is evaluated across this entire loading and unloading cycle. The percentage for each interval can be readily obtained by normalizing the vector by dividing the total weights. The total external and internal trip counts are set as $N_{12} = 6000$ (veh) and $N_{22} = 11000$ (veh), respectively. These trips counts are consistent among all tested scenarios.

TABLE 3 Summary of parameters for tested demands

Parameter	Tested Values
Sensitivity s	$2^0, 2^1, 2^2, 2^3, 2^4, 2^5, 2^6, 2^7$
Number of hops h	0, 2, 4, ..., 22 (increments of 2)
Asynchrony τ	0, $\frac{1}{4}$, $\frac{2}{4}$, $\frac{3}{4}$, 1 (hour)
Imbalance $\alpha_{22}^{\text{upper}}$	50%, 55%, 60%, 65%, 70%, 75%, 80%

4.3 | Compared baselines

We carefully selected 4 baselines that directly highlight the strengths of our approach. All baselines share the same first-stage deep RL controller for fair comparison. This design isolates the effect of the redistribution metric, so that any performance difference is attributable to the metric rather than to differences in learning architectures or training. For the same reason, we do not attempt to reproduce recent complex multi-agent deep-RL perimeter controllers as baselines: their code is generally unavailable and their many architecture and hyperparameter choices would confound this controlled comparison. Our metric is instead complementary to such methods, since it can be embedded as a module within them, which we regard as a future direction rather than viewing our methodology as a competing baseline.

1. *Homogeneous PC*²⁴: This pre-trained RL-based control policy shown in Fig. 3 operates more smoothly than the Bang-bang control that suffers from oscillation, and is more proactive and adaptive than the reactive PI feedback control.
2. *Heterogeneous PC with Myopic Pressure (Hetero: 2-hop)*: As a variant of our proposed method, this baseline represents a heterogeneous PC approach where the total permitted inflow is redistributed based on the vanilla myopic pressure. It serves as a straightforward comparison point to highlight the benefits of extending to multi-hop pressure.

3. *Queue Balancing (QB)*⁶: Inflow redistribution is framed as a quadratic programming knapsack problem to redistribute the total permitted inflow, where the objective is to balance queues on feeder links:

$$\min_{a_t} \sum_{f=1}^{|\mathcal{F}|} \frac{(Q_t(f) - C a_t^f)^2}{Q_{\max}(f)} \quad (21)$$

$$\text{s.t. } \sum_{f=1}^{|\mathcal{F}|} a_t^f = A_t \quad (\text{total inflow conservation}), \quad (22)$$

where $Q_{\max}(f)$ is the queue storage capacity of a feeder link f , and $C = 96$ s is the control action update cycle. A_t is the total inflow from the first stage control.

4. *Clustered N-MP*¹⁴: To consider a wider range of congestion, this baseline employs an *equal* weighting scheme on multi-hop downstream links (forming a cluster) using a *simple average* when the cluster's queue density exceeds a critical density ρ_{cr} . Hence, unlike our approach, neither influence decay over high hops nor turning ratios are integrated in Clustered N-MP. The calculation of Clustered N-MP for some link l with h -hop scope, denoted as $p_{\text{NMP}}(l, h)$, is defined in Eq. (23) where $\rho(l, h)$ is the average queue density of link l 's h -hop downstream links (cluster), and critical density $\rho_{\text{cr}} = 0.04$ veh/m/lane = 40 veh/km/lane. Hyperparameters ξ in Eq. (24) and χ in Eq. (25) controlling shape of the modified Sigmoid function are set to $\xi = 6$, $\chi = 400$. To align the experimental setup, we implement the Clustered N-MP style weighting scheme based on queue density. To the best of our tuning ability, we aimed to ensure that Clustered N-MP performed optimally as a comparative baseline against our method.

$$p_{\text{NMP}}(l, h) = \quad (23)$$

$$\begin{cases} p(l, 1) - \Psi(\rho(l, h), \rho_{\text{cr}}, Q(l)) & \text{if } \rho(l, h) > \rho_{\text{cr}} \\ p(l, 1) & \text{if } \rho(l, h) \leq \rho_{\text{cr}} \end{cases}$$

$$\Psi(\rho(l, h), \rho_{\text{cr}}, Q(l)) = \xi \cdot (\rho(l, h) - \rho_{\text{cr}})^2 \cdot \sigma(Q(l)) \quad (24)$$

$$\sigma(Q(l)) = \left(\frac{1}{1 + \exp(\frac{-Q(l)}{\chi})} - \frac{1}{2} \right) \times 10^3 \quad (25)$$

5 | RESULTS AND DISCUSSION

In this section, we analyze and discuss the results. For consistency, unless stated otherwise, all aggregate results reported in this section (including the baseline comparison and RQ1 to RQ3) are averaged over 10 random seeds to account for traffic stochasticity, while the turning-ratio robustness study in Section 6 uses 20 samples per perturbation level.

5.1 | Comparison with baselines

The results demonstrate that our 8-hop approach outperforms both the myopic version (2-hop) and the N-MP baseline across multiple metrics. Table 4 shows that our 8-hop method achieves the lowest total time spent (TTS), with significant improvements in the protected region, while slightly higher TTS outside compared to three baselines. Fig. 7a and Fig. 7b further highlight the superiority of our 8-hop approach in handling network-wide traffic. Both panels plot each method's signed difference relative to the homogeneous baseline over the simulation horizon, so a value is negative when a method temporarily performs worse than homogeneous control. Our 8-hop strategy achieves the highest cumulative trip completion rate (Fig. 7a) and the most substantial reduction in network density (Fig. 7b) relative to homogeneous control.

The Queue Balancing (QB) method, which redistributes inflow by minimizing queue imbalance across feeders, performs better than homogeneous control and the 2-hop method, as seen in Table 4. However, it still underperforms compared to our 8-hop method and N-MP. While QB ensures total inflow conservation, it lacks broader spatial awareness to consider downstream congestion beyond immediate neighbors. This limitation is reflected in its lower trip completion rate in Fig. 7a and higher network density in Fig. 7b. Overall, QB offers moderate improvement but is limited by its myopic optimization scope.

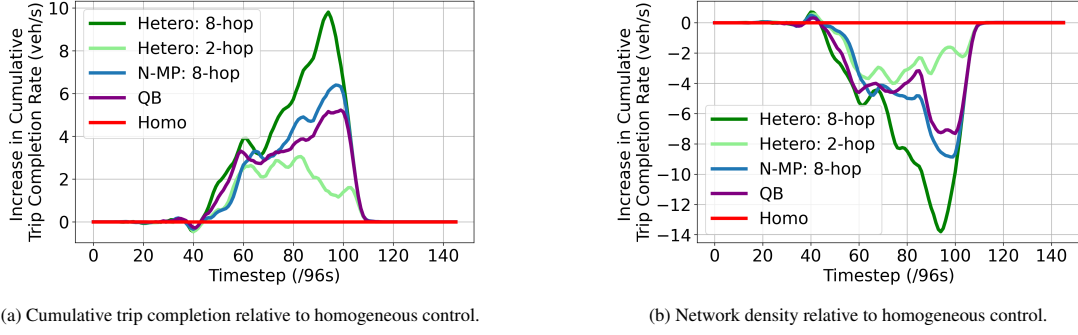


FIGURE 7 PC method time-series comparisons (differences relative to homogeneous control).

The performance of our 8-hop approach attributes to the incorporation of a broader spatial horizon, capturing the impact of multi-hop downstream traffic conditions more effectively. Unlike the myopic 2-hop method, which considers a limited spatial scope, and N-MP, which applies an equal weighting scheme without accounting for the decaying influence of higher hops, our 8-hop approach integrates multi-hop pressures by considering decaying influence of far downstream links with a mathematically grounded Markov chain framework. This allows a more informed redistribution of inflows.

It is worth explaining why, in Fig. 7a, the myopic 2-hop method can temporarily perform *worse* than the homogeneous baseline during the peak congestion period. A 2-hop metric only observes the immediate downstream. During the peak, a feeder link whose *near* downstream is relatively empty but whose *far* downstream is congested is misjudged as under-utilized, so the 2-hop controller admits more inflow there and aggravates the far-downstream congestion, thereby performing worse than uniform (homogeneous) admission. This failure mode is precisely the motivation for looking multiple hops downstream: our multi-hop metric perceives the distant congestion and avoids the misallocation, which is why it recovers and outperforms the homogeneous baseline.

TABLE 4 Comparison of TTS (hours); the last row gives the improvement relative to homogeneous control.

Method	Hetero: 8-hop	N-MP: 8-hop	Hetero: 2-hop	QB	Homo
TTS	2457	2686	2933	2710	3226
TTS Inside	1979	2447	2643	2545	2883
TTS Outside	478	239	290	165	343
Improv. vs Homo	23.8%	16.7%	9.1%	16.0%	0.0%

5.2 | RQ1: Calibration of control parameter s

Fig. 8 shows the TTS for the heterogeneous controller across different values of pressure sensitivity s . The results, averaged over simulations with 10 random seeds to account for traffic stochasticity, indicate that the performance improves significantly when s is within an optimal range between 2^3 and 2^4 . Outside this range, performance drops, as both overly low and excessively high values of s negatively impact the controller's effectiveness. This highlights the importance of calibrating the control parameters to optimize performance.

5.3 | RQ2: Optimal hops and performance convergence

The TTS results in Fig. 8 also reveal that performance improves as the number of hops increases, showing a trend of gradual enhancement compared to the baseline of homogeneous control (labeled 'Homo'). The performance converges after approximately 10 hops, indicating that beyond this range, there is minimal additional benefit. One might expect that, given the geometric size of the network, more than about six hops would be unnecessary. However, vehicles do not travel only along shortest paths: turning

movements create *tortuous* walks, so downstream influence propagates beyond the straight-line block distance. This is why marginal improvements persist slightly past eight hops. These contributions nevertheless vanish quickly because the turning-probability-weighted influence decays geometrically with hop (see the boundedness argument in Section 3.5 and Appendix B), which is what produces the convergence beyond about ten hops.

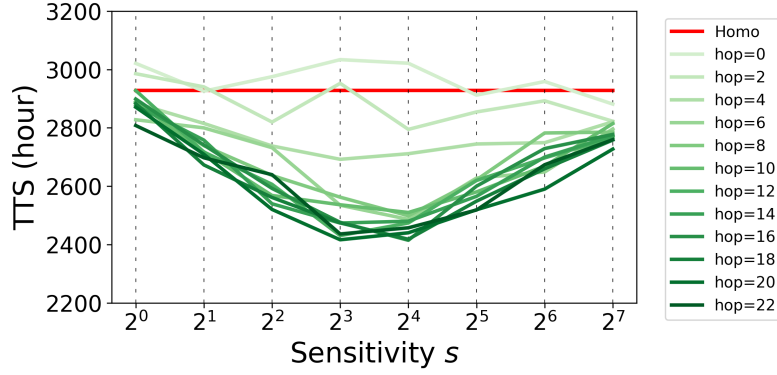


FIGURE 8 TTS vs. sensitivity for the heterogeneous controller.

5.4 | Decayed impact of downstream links over hops

The performance convergence can be further explained by the decaying influence over hops. Fig. 9 visualizes the 20-hop downstream link accumulative importance for feeder link 5. The accumulative importance reflects how traffic conditions from 1-hop to 20-hop downstream links impact the upstream traffic at feeder link 5. A darker color of a downstream link indicates a greater impact on feeder link 5. A distinct decay in importance with increasing hop is observed, emphasizing a trend where proximal downstream links exert a more pronounced effect on the feeder link than those distant downstream links. The decay pattern provides insight into the performance convergence beyond 10 hops: as the importance of downstream links diminishes over hop, the incremental contribution beyond 10 hops becomes increasingly negligible. This analysis validates the importance of downstream links in proximity and the rationale behind employing a finite spatial scope in multi-hop pressure.

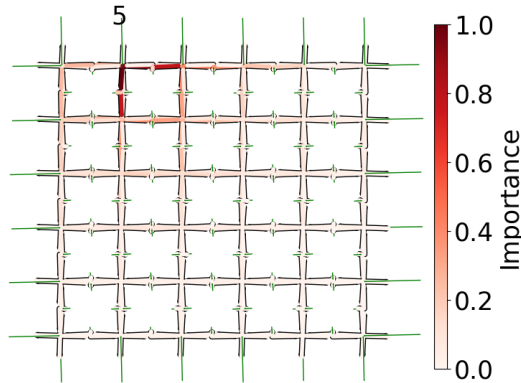


FIGURE 9 Heatmap of the accumulated importance of 20-hop downstream links for feeder link 5.

5.5 | Profile of pressures for each feeder link

In Fig. 10, we compare the traffic pressure profiles under homogeneous control and heterogeneous control. Both left and right figures capture the pressure at different hops for each feeder link at the same timestep. Under homogeneous control, the pressure profile on the left figure indicates a relatively imbalanced distribution across the feeder links, especially upper feeder links labeled in blue and lower feeder links labeled in red. This is because the homogeneous control cannot differentiate among the feeder links' varying congestion levels. The negative pressure with a large magnitude for lower feeder links indicates very congested downstream conditions. Conversely, the heterogeneous control demonstrates an equalized pressure distribution across feeder links. The small magnitude of pressures indicates the feeder link and its downstream have similar traffic potentials, and the downstream is not too congested. The heterogeneous controller modulates the total permitted inflow more adaptively by allocating higher permitted inflow to a feeder link with larger pressure (possibly an uncongested downstream) and reducing the permitted inflow to a feeder link with smaller pressure (possibly a congested downstream), potentially alleviating congestion in both feeder links and downstream.

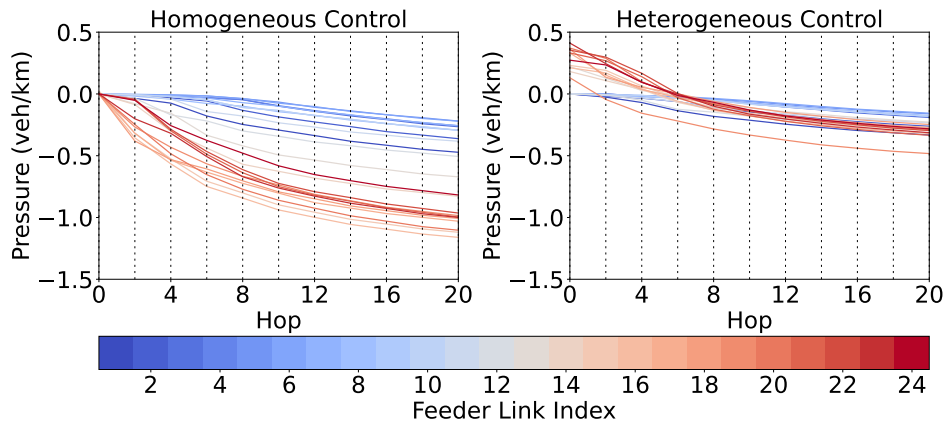


FIGURE 10 Pressure profiles of the feeder links under homogeneous and heterogeneous control.

5.6 | RQ3: Performance vs demand heterogeneity

Fig. 11 shows a heatmap of how the performance improvement relates to two sources of demand heterogeneity: the demand shift τ (asynchrony) and the upper subregion internal demand percentage $\alpha_{22}^{\text{upper}}$ (imbalance). The demand setup on the top right is the most heterogeneous, while the bottom left is the most homogeneous. The gradient of colors, ranging from green to red, conveys the magnitude of improvement, with more intense greens indicating higher performance improvement. Generally speaking, an increase in demand heterogeneity correlates with a more pronounced improvement in performance. Conversely, demand setup with lower heterogeneity exhibits minimal or slightly negative performance improvement denoted by the light green and red cells. This trend confirms that the proposed heterogeneous controller is more effective in a more heterogeneous scenario (i.e., as we progress up and right in Fig. 11).

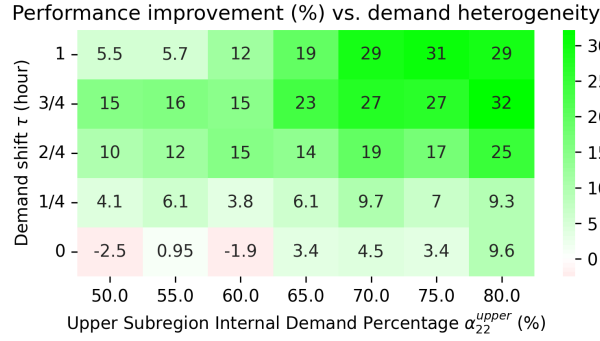


FIGURE 11 Performance improvement (%) of the heterogeneous controller relative to homogeneous control across the demand-heterogeneity grid (asynchrony τ vs. imbalance $\alpha_{22}^{\text{upper}}$).

6 | SENSITIVITY ANALYSIS: ROBUSTNESS AGAINST TURNING RATIO PERTURBATIONS

6.1 | RQ4: Robustness against turning ratio perturbation: Is the heterogeneous PC resilient to uncertainties in turning ratio estimations?

The objective of this section is to assess the robustness of the multi-hop pressure-based approach when faced with turning ratio perturbations. Note that baseline methods like Homogeneous control, Queue Balancing (QB), and Clustered N-MP do not incorporate turning ratios, therefore they are not part of the robustness test.

6.2 | Experimental setup

The experimental design involves sampling turning ratios for each link in the traffic network. For each traffic simulation, the turning ratio of each link is sampled according to a probabilistic distribution and fixed during that simulation. The simulation is repeated 20 times for each distribution at a specified perturbation level to gather statistical data. To ensure the only random variable is the turning ratio, the simulation seed is fixed to 1. For these controlled experiments we assume perfect queue-density information, so that turning-ratio uncertainty is isolated as the single studied perturbation. We note that the metric is also inherently cushioned against queue-density estimation error: distant links enter the pressure with rapidly decaying weights, so their estimation noise is heavily attenuated (the dominant terms being the well-observed near links), and the metric is bounded (Appendix B), preventing error amplification. A dedicated queue-density robustness study is left as future work.

6.3 | Turning ratio perturbations

The perturbed turning ratios are sampled from a Dirichlet distribution (See Eq. (6.3)) in which its sample naturally satisfies the property of probability space. The mean of the distribution is parameterized by a weighted average of the original turning ratios $\mathbf{T}_i = \bigoplus_{j \in \mathcal{N}(i,1)} T_{ij}$ and a uniform vector $\mathbf{1}_i = \bigoplus_{j \in \mathcal{N}(i,1)} 1$. The parameter $M = 100$ is the inverse temperature that regulates the variance of the Dirichlet distribution.

$$\mathbf{T}_i \sim \text{Dir}(M(1 - \alpha)\mathbf{T}_i + M\alpha\mathbf{1}_i) \quad (26)$$

The degree of perturbation is controlled by the parameter $\alpha \in [0, 1]$, where $\alpha \rightarrow 0$ implies minimal perturbation and $\alpha \rightarrow 1$ signifies maximal perturbation. Notably, a setting of $\alpha = 1$ results in the expectation of equal turning ratios, e.g., $[\frac{1}{3}, \frac{1}{3}, \frac{1}{3}]$ for a link with three turning movements. To clarify, the minimal perturbation ($\alpha = 0$) does not necessarily signify no perturbation, but rather results in a probabilistic distribution, where the mean of the distribution is the original turning ratio.

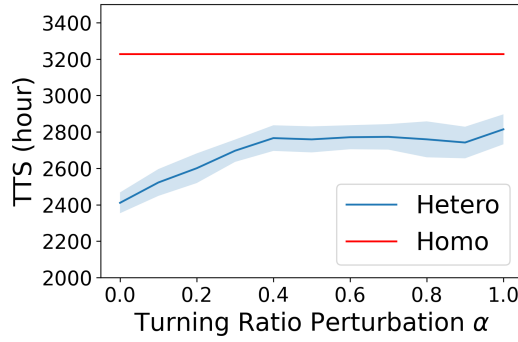


FIGURE 12 Robustness of the heterogeneous controller against turning-ratio perturbations.

6.4 | Results of sensitivity analysis

In this part, we show the result of the robustness of the heterogeneous controller against turning ratio perturbations. Fig. 12 depicts performance degradation as perturbations intensify. Despite this, even at the highest perturbation level, where the mean of the Dirichlet distribution is equal turning ratios (e.g., $1/3$ for a link with 3 turns), the heterogeneous controller still outperforms the homogeneous control. It suggests that while the heterogeneous controller is affected by turning ratio perturbations, its adaptive mechanism still provides a superior performance in managing traffic flow under uncertainty. Therefore, the sensitivity analysis demonstrates a strong robustness of our proposed approach. Note that there is no variability for homogeneous control in Fig. 12 because we fixed the tested scenario to ensure the only source of uncertainty is turning ratio perturbation.

7 | CONCLUSION AND DISCUSSION

This paper investigates the generalization of multi-hop traffic pressure for heterogeneous perimeter control. Our formulation of multi-hop traffic pressure provides insight not only into immediate downstream link conditions but also into more distant traffic conditions. Our customizable long-distance traffic metric, grounded in Markov chain theory, bridges the gap between MFDs and 1-hop pressure where the spatial granularity is either too coarse or too fine. To account for the greater influence of closer traffic links, multi-hop pressure naturally discounts distant downstream links by the product of turning ratios, interpreted as the multi-step transition probability in the Markov chain. This novel metric, supported by a clean mathematical formulation, effectively guides heterogeneous perimeter control: in brief, the total permitted inflow is produced by a pre-trained macroscopic controller reused from prior work, and our contribution is the multi-hop downstream pressure metric together with its use as a *modular inflow-redistribution mechanism*. Because the metric operates on any total-inflow signal, it can be paired with different perimeter-control architectures rather than constituting a wholesale new controller.

Analysis of results provides insights into the multi-hop pressure based on the heterogeneous controller. Within the optimal range of pressure sensitivity, the controller performs better than when the pressure sensitivity is outside of this range, which emphasizes the importance of control parameter calibration. In addition, the convergence of performance as hops increases indicates that knowing extremely distant traffic conditions does not contribute to performance improvement. Such convergence can be explained by the decay of downstream link importance over large hops. Furthermore, our result shows that performance improvement is greater when the demand heterogeneity is larger, whereas under the most homogeneous demand, the improvement is minimal since heterogeneous perimeter control extensions are unnecessary in this case. A sensitivity analysis demonstrates robustness against turning ratio perturbation (e.g., arising in practice due to uncertainty in turning ratio estimation).

In real-world traffic systems, the turning ratios can be estimated from historical traffic data, while real-time sensors like loop detectors and cameras enable dynamic updates based on observed vehicle movements. Similarly, the estimation of queue densities needed for the multi-hop pressure computation can also benefit from modern traffic sensors such as cameras.

In terms of related work, our inflow-redistribution mechanism is related to, but distinct from, queue-balancing redistribution⁶: whereas queue balancing redistributes inflow to equalize feeder queues and is primarily concerned with conditions and effects *outside* the protected region, our metric targets the *internal* downstream congestion inside the region. These two perspectives are

complementary and could operate simultaneously, managing internal heterogeneity while also bounding external queues, which we regard as an attractive combined direction.

In addition, it is also important to note that this paper focuses on how to heterogeneously control the perimeter inflows given a *fixed* perimeter, rather than dynamic perimeter identification (DPI). Nonetheless, our methodology remains applicable even if the perimeter is dynamically adjusted. The problem of DPI is orthogonal to our current work and would require a separate investigation. Additionally, infrastructure constraints and practical considerations may often limit the feasibility of frequent perimeter change, making the fixed-perimeter approach evaluated in this work a practical solution.

Future work could 1) further generalize the pressure definition by extending the upstream potential to multiple hops; 2) incorporate dynamic turning ratio estimation in multi-pressure calculation; 3) formulate other heterogeneous perimeter controllers; 4) test on a more complex city network; 5) estimate queue densities under uncertainty; and 6) integrate DPI with our approach, as well as adaptive traffic signal control instead of flow meterings; 7) evaluate the approach under incident scenarios, where an incident-induced downstream queue would raise the corresponding negative pressure and automatically throttle the affected feeder; 8) conduct a dedicated robustness study to queue-density estimation error, analogous to the turning-ratio sensitivity analysis; and 9) embed the (differentiable) multi-hop pressure computation as a module in an end-to-end trainable controller, trading the current interpretability and training-free operation for potentially learned weightings.

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APPENDIX

A SCALAR VERSION: MULTI-HOP DOWNSTREAM PRESSURE FOR A SINGLE LINK

To gently guide the reader step-by-step, we first review the standard 1-hop traffic pressure, then demonstrate how to extend the 1-hop pressure to 2-hop and higher-hop versions.

A.1 1-hop Pressure

Adopted from physics principles, the existing standard 1-hop traffic pressure⁴⁰ is defined as the difference between upstream queue density and the summation of 1-hop downstream queue densities weighted by turning ratios. Mathematically, for a link l , its 1-hop pressure is:

$$p(l, 1) = Q(l) - \sum_{j \in \mathcal{N}(l, 1)} T_{lj} Q(j) \quad (\text{A1})$$

Apparently, 1-hop pressure is myopic in terms of knowing the traffic conditions further down the link of interest. When considering further neighborhoods, the concept of 1-hop pressure can be extended to multi-hop pressure to account for a wider range of traffic networks, capturing the cumulative effect of traffic congestion in neighboring areas.

A.2 2-hop Pressure

As per the requirement for multi-hop pressure, the congestion at 2-hop downstream links has less influence, and is naturally discounted by the turning ratio from link l to 1-hop downstream links:

$$p(l, 2) = Q(l) - \sum_{i_1 \in \mathcal{N}(l, 1)} T_{li_1} \left(Q(i_1) + \sum_{i_2 \in \mathcal{N}(i_1, 1)} T_{i_1 i_2} Q(i_2) \right) \quad (\text{A2})$$

$$= Q(l) - \underbrace{\sum_{i_1 \in \mathcal{N}(l, 1)} T_{li_1} Q(i_1)}_{p(l, 1)} - \sum_{i_1 \in \mathcal{N}(l, 1)} \sum_{i_2 \in \mathcal{N}(i_1, 1)} T_{li_1} T_{i_1 i_2} Q(i_2) \quad (\text{A3})$$

$$= p(l, 1) - \sum_{i_1 \in \mathcal{N}(l, 1)} \sum_{i_2 \in \mathcal{N}(i_1, 1)} T_{li_1} T_{i_1 i_2} Q(i_2) \quad (\text{A4})$$

A.3 Multi-hop Pressure

As one may notice, there exists a recursive relationship between two consecutive hops of pressure. The second term represents the additional pressure exerted on link l due to congestion h hops away that was not accounted for in $(h - 1)$ -hop pressure. By subtracting the second term, we isolate the pressure that is specifically due to the h -hop congestion, distinguishing it from the cumulative pressure calculated up to $h - 1$ hops:

$$p(l, h) = p(l, h - 1) - \sum_{i_1 \in \mathcal{N}(l, 1)} \sum_{i_2 \in \mathcal{N}(i_1, 1)} \cdots \sum_{i_h \in \mathcal{N}(i_{h-1}, 1)} T_{li_1} T_{i_1 i_2} \cdots T_{i_{h-1} i_h} Q(i_h) \quad (\text{A5})$$

Explained in graph walk and Markov chain, the product of h turning ratios in the second term is the probability of graph walk with exactly h edges that lead from vertex l to vertex i_h , and this probability measures the h -step influence of h -hop downstream link i_h towards link l . The summation in the second term accumulates all the influence of h -hop downstream links on traffic link l .

B PROPERTIES OF MULTI-HOP DOWNSTREAM PRESSURE

The multi-hop downstream pressure has two properties:

1. **Monotonously decreasing over the number of hops h :**

$$p(l, h) \leq p(l, h - 1) \quad \forall l \quad (\text{B6})$$

2. **Bounded range** (Assuming queue density is normalized to $[0, 1]$ by dividing $Q_{\max}$):

$$p(l, h) \in [-h, 1] \quad \forall l \quad (\text{B7})$$

The upper bound is reached when the link itself is full and all its downstream links do not have queued vehicles. In contrast, the lower bound is reached when the link itself does not have any queues while all downstream links are full.

Proposition: Lower bound of h -hop pressure is $-h$:

Proof. **Base case** ($h = 1$): Using the property that the normalized queue density satisfies $0 \leq Q(l) \leq 1$, we have

$$p(l, 1) = Q(l) - \sum_{j \in \mathcal{N}(l, 1)} T_{lj} Q(j) \quad (\text{B8})$$

$$\geq 0 - \sum_{j \in \mathcal{N}(l, 1)} T_{lj} \times 1 \quad (\text{B9})$$

$$= -1 \quad (\text{B10})$$

Hence, it establishes the base case.

Inductive step: Assume the proposition holds for $h - 1$, i.e., $p(l, h - 1) \geq -(h - 1)$. For h , the following holds:

$$p(l, h) = p(l, h - 1) - \sum_{i_1 \in \mathcal{N}(l, 1)} \sum_{i_2 \in \mathcal{N}(i_1, 1)} \cdots \sum_{i_h \in \mathcal{N}(i_{h-1}, 1)} T_{li_1} T_{i_1 i_2} \cdots T_{i_{h-1} i_h} Q(i_h) \quad (\text{B11})$$

$$\geq -(h - 1) - \sum_{i_1 \in \mathcal{N}(l, 1)} \sum_{i_2 \in \mathcal{N}(i_1, 1)} \cdots \sum_{i_h \in \mathcal{N}(i_{h-1}, 1)} T_{li_1} T_{i_1 i_2} \cdots T_{i_{h-1} i_h} \times 1 \quad (\text{B12})$$

$$= -(h - 1) - \sum_{i_1 \in \mathcal{N}(l, 1)} T_{li_1} \sum_{i_2 \in \mathcal{N}(i_1, 1)} T_{i_1 i_2} \cdots \sum_{i_h \in \mathcal{N}(i_{h-1}, 1)} T_{i_{h-1} i_h} \quad (\text{B13})$$

$$= -(h - 1) - 1 \times 1 \cdots \times 1 \quad (\text{B14})$$

$$= -(h - 1) - 1 \quad (\text{B15})$$

$$= -h \quad (\text{B16})$$

Thus, by the principle of induction, the lower bound of h -hop pressure is $-h$ for all $h \geq 1$. $\square$

CONFLICT OF INTEREST

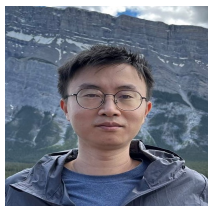
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